

Brothers Mobile Shop ERP

Complete System Architecture, REST API Reference & Comprehensive Client-Side Manual

REPOSITORY

Omegalflop-Intern /
ERP_Management

ARCHITECTURE

Node 20 + React 18 Vite +
MongoDB 7

ENTERPRISE MODULES

26 Full Stack Modules

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1 Executive System Overview

Brothers Mobile Shop ERP is an enterprise platform engineered specifically for retail and wholesale mobile phone businesses. It unifies Point of Sale (POS), real-time IMEI serial tracking, multi-branch stock transfers, double-entry financial accounting, HR payroll, warranty management, and technician device repairs into an integrated system with live Server-Sent Events (SSE).

System Access & Infrastructure URLs

Client Web App: `http://localhost:3000`
(React 18 + Vite)

Backend REST API:
`http://localhost:5000/api/v1` (Node
Express)

Interactive Swagger API:
`http://localhost:5000/api/docs`

MongoDB Connection:
`mongodb://127.0.0.1:27017/mobile_shop_erp`

Default Access Credentials

ROLE	USERNAME	PASSWORD SETUP
System Administrator	admin	SEED_PASSWORD_ADMIN in server/.env
Branch Manager	manager	SEED_PASSWORD_MANAGER in server/.env

2 How to Run & Access the System

Development Mode (Separate Terminals)

Terminal 1: Start Backend API Server (Port 5000)

```
cd server && npm install && npm run seed && npm run dev
```

Terminal 2: Start Frontend Client (Port 3000)

```
cd client && npm install && npm run dev
```

Docker & Production Deployment

Automated multi-container launch (MongoDB + Server + Client):

```
docker compose up -d --build
```

Production PM2 Server Launch:

```
cd server && npm start
```

3 Core Business Workflows & Data Flow

Workflow A: Sales & IMEI Stock Deduction

- 1 Cashier opens POS at `/sales/new` & scans IMEI/barcode.
- 2 Client verifies item status is `In Stock` in local cache.
- 3 Order processes; generates invoice, updates IMEI to `Sold`, posts ledger.

Workflow B: Device Repair Management

- 1 Technician logs incoming device on `/repairs` with issue diagnosis.
- 2 Pipeline updates: `Received` → `In Progress` → `Repaired` → `Delivered`.
- 3 Upon delivery, repair invoice generates & links to double-entry revenue.

Workflow C: HR Attendance & Payroll

- 1 Staff logs daily check-in / check-out on `/hr/attendance`.
- 2 System computes overtime hours & unpaid leave deductions.
- 3 Payroll generates salary sheet, printable payslips & posts expense.

4 Complete Client-Side Web Application Architecture & Module Map

Comprehensive page-by-page operational map documenting all client-side page components, routes, user actions, forms, modals, state management, and RBAC permissions.

1. Authentication & Onboarding Modules

`/login` — Login Page `Public Route`

- **Capabilities:** Supports direct credential login (`/auth/login-direct`) for development speed, OTP email verification flow toggle, and password visibility toggle.
- **UI Controls:** Identifier input, Password input, Remember Me checkbox, Design Theme selector widget, Login submit button with loading spinner state.
- **State Management:** Uses `useAuth()` context to set JWT token, user profile, role permissions, and persist token to `localStorage`.

`/forgot-password` & `/verify-email` `Public Route`

- **Capabilities:** Email password recovery request form, OTP code entry input, email verification link handler, and instant toast notifications.

2. Executive Dashboard (`/dashboard`)

`/dashboard` — Operations Command Center `Protected Route`

- **KPI Metrics Cards:** 4 live metric cards displaying Today's Total Sales Amount, Low Stock Alert Count, Active Device Repairs, and Pending Customer Dues with percentage trend indicators.
- **Analytics Charts:** Interactive sales revenue line chart (7-day / 30-day view) and product category revenue breakdown donut chart built with Recharts.
- **Recent Activity Feed:** Live transaction stream table showing recent sales, purchases, and repair status changes updated in real-time via Server-Sent Events (SSE).
- **Quick Action Hub:** One-click launch buttons for New POS Sale, Add Product, Register Repair Job Sheet, and Log Operational Expense.

3. Point of Sale (POS) & Sales Management

`/sales/new` — POS Checkout Terminal `Permission: sales:create`

- **High-Speed Scanner Input:** Dedicated barcode / IMEI serial number scanner input field with auto-focus and instant product lookup.
- **Sales Cart Manager:** Cart table displaying items, unit prices, custom discounts, IMEI selector modal for serial-tracked items, and quantity modifiers.
- **Customer Selector & Quick Add:** Instant customer search by phone/name, quick creation modal for walk-in customers, and due balance display.
- **Multi-Channel Payment Modal:** Split payment handling supporting Cash, Credit Card, Mobile Banking (bKash/Nagad), and Customer Credit (Due), with automatic change calculation.
- **Thermal Receipt Engine:** Generates instant printable receipts optimized for `80mm` thermal printers, complete with store header, item barcode, and QR invoice verification code.

`/sales` — Sales History & Invoices `Permission: sales:view`

- **Filterable Datatable:** Comprehensive invoice table with filters for Date Range, Branch, Payment Status (Paid, Partial, Due), and Search by Invoice Number/Customer.
- **Actions:** View Detailed Invoice, Reprint Thermal Receipt, Download Invoice PDF, and Initiate Sales Return.

`/sales/returns` — Sales Returns & Refunds `Permission: sales:create`

- **Return Workflow:** Look up original invoice, select return items/IMEIs, verify restock condition, calculate refund/credit adjustment, and update accounting ledger automatically.

4. Inventory, Stock & IMEI Tracking Modules

`/products` — Master Product Catalog `Permission: products:view`

- **Master Product Grid & Table:** View items with thumbnail image, SKU, Brand, Category, Purchase Price, Selling Price, Cost Margin %, and current total stock badge.
- **Product Form Modal:** Add/Edit product modal supporting image upload, barcode assignment, low-stock threshold alert setting, and IMEI serial tracking toggle.

`/inventory` — IMEI Serial Number Tracker `Permission: inventory:view`

- **Serial Lifecycle Tracker:** Search any IMEI number to view its status badge (`In Stock` , `Sold` , `In Repair` , `Transferred` , `Returned`), branch location, purchase invoice link, and sales invoice link.

`/stock` & `/stock-transfer` — Stock Operations `Permission: stock:view, stock:transfer`

- **Branch Stock Matrix:** Overview of inventory across all store branches with stock valuation totals.
- **Inter-Branch Stock Transfer:** Transfer request creation, branch selection, IMEI selection, transfer approval workflow, and destination barcode scan intake.

5. Device Repair & Service Operations

/repairs — Device Repair Hub Permission: repairs:view, repairs:manage

- **Service Job Sheet Generator:** Intake modal recording customer device details (Brand, Model, IMEI/Serial, Passcode, Condition), reported faults, initial cost estimate, and assigned technician.
- **Kanban / Pipeline View:** Visual drag-and-drop or dropdown status tracking: Received → Diagnosing → In Progress → Repaired → Delivered .
- **Parts & Labor Billing:** Add repair spare parts used from inventory and technician labor charges to repair job sheet.
- **Repair Delivery & Billing:** Generates repair invoice upon delivery, updates customer account, and posts service revenue to double-entry accounting.

6. Customer Relationship Management (CRM) & Warranties

`/customers` & `/customers/:id` — **Customer Directory** Permission: `customers:view`

- **Customer Directory:** List of retail and wholesale customers showing total purchases, outstanding due balance, and phone/email contacts.
- **Customer Profile View:** Complete history of purchases, payment receipts, due ledger, and active warranty claims.

`/customers/due-collection` — **Due Payment Collection** Permission: `customers:view`

- **Due Settlement Interface:** Quick collection input for customer outstanding balances, partial payment support, receipt printing, and payment channel logging.

`/warranties` & `/warranties/report` — **Warranty Claims** Permission: `warranties:view`

- **Warranty Serial Check:** Instantly check warranty validity by scanning IMEI number. Log replacement/repair claim with supplier return tracking.

7. Purchases & Supplier Management

`/suppliers` & `/purchases` — **Supplier & Purchase Intake** Permission: `purchases:view`, `suppliers:view`

- **Supplier Directory:** Vendor master list, contact info, total orders placed, and payable supplier dues balance.
- **Goods Received Note (GRN) & Purchase Form:** Log incoming stock shipments, cost prices, bulk IMEI serial upload via CSV/text, auto-update stock quantities, and generate purchase invoice.

8. Double-Entry Financial Accounting Modules

Chart of Accounts & Journal Entries Permission: `accounting:view`

- `/accounting` — Interactive tree view of Assets, Liabilities, Equity, Revenue, and Expense accounts. Add custom sub-accounts.
- `/accounting/journal-entries` — Manual multi-line Debit/Credit journal voucher creation with mandatory balance verification.

Financial Statements & Accounting Pages Permission: `accounting:view`

- `/accounting/balance-sheet` — Real-time Balance Sheet statement (Assets = Liabilities + Equity).
- `/accounting/profit-loss` — Income statement breakdown of Gross Revenue, COGS, Operating Expenses, and Net Profit.
- `/accounting/trial-balance` — Trial balance report verifying debit/credit ledger parity.
- `/accounting/expenses` — Operational expense logging with category classification and voucher attachment upload.
- `/accounting/investors` — Investor capital tracking, profit-sharing distribution calculation, and dividend payout history.
- `/accounting/assets` & `/accounting/loans` — Fixed asset depreciation schedules and business loan repayment trackers.

9. HR, Attendance & Payroll Modules

Staff Management, Attendance & Leaves Permission: employees:view, attendance:view

- `/hr/employees` — Staff profiles, designations, basic salary configuration, join date, and document attachments.
- `/hr/attendance` — Daily check-in/out terminal, working hours computation, late check-in tracking, and monthly attendance summary.
- `/hr/leaves` — Leave application form, leave balance tracker, and multi-level manager approval/rejection controls.

Automated Payroll Sheet `/hr/payroll` Permission: payroll:view

- **Monthly Salary Calculation:** Automated calculation of gross salary (Basic Salary + Overtime Hours - Unpaid Leave Deductions - Penalties).
- **Salary Slips:** One-click generation of printable salary slips and automatic salary expense posting to accounting.

10. Administration, Security & System Settings

User Accounts, RBAC Roles & Audit Trail Permission: users:manage, roles:manage

- `/users` — System user management table, branch restriction assignment, activate/deactivate account toggle, password reset modal.
- `/roles` — Role-Based Access Control (RBAC) permission grid allowing granular toggling of 26 module permissions.
- `/settings/activity-logs` — System audit log table tracking user actions, IP addresses, login timestamps, and security alert events (e.g. `BRUTE_FORCE_DETECTED`).

Enterprise System Settings `/settings` Permission: settings:manage

- **Company Profile:** Business name, logo upload, phone, tax ID, and address configuration for invoices.
- **Thermal Receipt Setup:** Printer paper selection (80mm vs 58mm), header text, footer return policy text, and logo toggle.
- **SMTP & System Backup:** Email server testing setup and manual database backup trigger button.

5 Complete REST API Endpoints Catalog (26 Modules)

API Base Route: `http://localhost:5000/api/v1`

Module	Base Route	Key REST Operations & Capability Description
Auth	<code>/auth</code>	Login (/login-direct), dev bypass, OTP email verification, Password reset, Logout.
User & Roles	<code>/users</code> , <code>/roles</code>	User CRUD, branch assignment, RBAC permission matrix management.
Catalog	<code>/catalog</code>	Categories, Brands, Models, Unit of Measurements (UOM) management.
Product	<code>/products</code>	Master product catalog, pricing structures, cost margin analysis, image uploads.
IMEI Tracker	<code>/imei</code>	Serial number lifecycle tracking (In Stock, Sold, Repaired, Transferred), full history.
Stock & Inventory	<code>/stock</code>	Stock quantities, low-stock threshold alerts, inter-branch stock transfers.
Supplier	<code>/suppliers</code>	Supplier profiles, purchase ledgers, outstanding dues, payment processing.
Purchase	<code>/purchases</code>	Purchase orders, incoming GRN intake, CSV bulk IMEI registration, purchase invoices.
Sale (POS)	<code>/sales</code>	Retail POS checkout, invoice PDF generation, thermal print receipts, customer returns.
Wholesale	<code>/wholesale</code>	Bulk B2B orders, volume pricing tiers, credit limit validation, wholesale invoices.
Customer (CRM)	<code>/customers</code>	Customer profiles, transaction ledger, due collection entries, purchase history.
Warranty	<code>/warranties</code>	IMEI warranty claims processing, policy check, replacement issue tracking.
Repair Service	<code>/repairs</code>	Repair job sheets, technician assignment, status pipeline, service invoices.
Accounting	<code>/accounting</code>	Chart of Accounts, journal vouchers, Balance Sheet, Profit & Loss, Trial Balance.
Expense	<code>/expenses</code>	Operational expense logging, expense categories, receipt attachment vouchers.
Investor	<code>/investors</code>	Investor capital tracking, profit-sharing distribution share calculation.
Loan	<code>/loans</code>	Business loan records, interest rate computation, EMI repayment schedule.
Employee	<code>/employees</code>	Staff profiles, designations, basic salary structures, document management.
Attendance	<code>/attendance</code>	Daily check-in/check-out logs, working hours computation, late arrival marks.
Leave	<code>/leaves</code>	Leave application submission, approval workflows, leave balance tracker.
Payroll	<code>/payroll</code>	Monthly salary sheet generation, overtime/bonus additions, printable pay slips.
Branch	<code>/branches</code>	Multi-branch location setup, manager assignment, branch sales analytics.
Settings	<code>/settings</code>	Company profile, thermal print templates, design themes, SMTP email setup.
Notification	<code>/notifications</code>	Real-time alerts, low-stock warnings, system audit security events.
Real-time SSE	<code>/sse</code>	Server-Sent Events live stream for instant POS notifications & inventory sync.

MODULE	BASE ROUTE	KEY REST OPERATIONS & CAPABILITY DESCRIPTION
Report Analytics	/reports	Revenue charts, sales analytics, top selling models, HR financial summaries.

6

Backup, Security & Disaster Recovery

Database Restoration & Backup Trigger

To execute automated or manual system restoration from backend backup files:

```
cd server && node scripts/run-restore.js
```

Security Architecture: Automatic brute-force IP rate-limiting (5 failures in 15 mins triggers `BRUTE_FORCE_DETECTED` audit log), JWT bearer authentication, Helmet HTTP security headers, strict MIME type & file magic number verification on uploads.